

MARKING SCHEME

Initial Assessment Test – Informatics and ICT, Grade IX

School year 2026-2027

Total: 100 points (10 points awarded by default + 90 points for the items). The grade is the total divided by 10.

Item 1 (10 points)

Answer: d) 10 numbers – 9, 18, 27, 36, 45, 54, 63, 72, 81, 90 (99 has a digit sum of 18, so it is excluded).

Criterion	Points
Correct choice (d)	5
All 10 numbers listed (0.5 points per correct number; deduct 0.5 points for each wrong number listed)	5

Item 2 (10 points) – 2.5 points per statement

Statement	Answer	Explanation
1	A	The "cat_timp" / "repeat until" loop is a pretest loop.
2	A	$n \leftarrow [n/10]$ is an assignment.
3	F	There is no input (read) operation.
4	F	All the digits of n are displayed, in reverse order.

Item 3 (10 points)

Model solution:

```
citeste x, y
a ← (x / 100) % 10      { 3rd digit of x }
b ← (y / 1000) % 10     { 4th digit of y }
c ← (y / 10) % 10       { 2nd digit of y }
n ← a*100 + b*10 + c
daca n % 7 = 0 atunci scrie "DA"
altfel scrie "NU"
```

Criterion	Points
Reading x and y	1
Extracting each of the three digits (2 points each)	6
Forming the number in the correct order	1.5
Divisibility test by 7	1
Displaying both messages (DA / NU)	0.5

Any correct method is accepted (for example, loops that strip digits, or different variable names).

Item 4 (10 points)

Input data: g1, m1, s1, g2, m2, s2. **Output data:** g, m, s (the sum, in degrees, minutes, seconds).

Model solution:

```

citeste g1, m1, s1, g2, m2, s2
s ← s1 + s2
m ← m1 + m2 + s / 60
s ← s % 60
g ← g1 + g2 + m / 60
m ← m % 60
scrie g, " ", m, " ", s

```

Check: 30 40 56 + 82 35 24 gives 113 16 20.

Criterion	Points
Input data identified correctly	2
Output data identified correctly	2
Reading and writing (with spaces as separators)	1
Seconds: sum and carry to minutes	2
Minutes: sum, carry received, carry to degrees	2
Degrees: sum and carry received (in the correct order of operations)	1

Item 5 (10 points)

Answer: $x \% 2 = 0$ și $x < y$. Equivalent forms are accepted, such as $x \% 2 == 0$ && $y > x$, $x <= y$, or " $x \bmod 2 = 0$ " and " $x < y$ " in the Scratch "and" block.

Criterion	Points
Correct condition $x \% 2 = 0$ (x is even)	4
Correct condition $x < y$ (x is smaller than the current minimum)	4
Conditions joined correctly with AND	2

Item 6 (15 points)

Find the smallest k such that $1 + 2 + \dots + k \geq n$.

Model solution:

```

citeste n
s ← 0
k ← 0
cat timp s < n executa
{ k ← k + 1
  s ← s + k }
scrie k

```

Check: $n = 10$ gives 4 days.

Criterion	Points
Reading n	1
Correct initialisations (s , k)	3
Loop with correct stopping condition ($s < n$, or equivalent)	4
Correct update of the day counter	2

Criterion	Points
Correct accumulation of the distance	3
Displaying the result	2

- An off-by-one error in the condition (such as $s \leq n$) loses the 4 points for the condition.
- A correct solution using a formula or another valid algorithm receives full marks.

Item 7 (15 points)

Solution: the maximum number of packages is $\text{gcd}(320, 240, 200) = 40$.

Each package contains $320/40 = 8$ **nuts**, $240/40 = 6$ **chocolates**, and $200/40 = 5$ **oranges**.

Criterion	Points
Recognising that the problem requires the GCD	5
Correct computation of the GCD (prime factorisation or Euclid's algorithm) = 40	4
Nuts per package (8)	2
Chocolates per package (6)	2
Oranges per package (5)	2

Item 8 (10 points)

Answer: a) the sum of the odd natural numbers less than or equal to n (k takes the values 1, 3, 5, ... while $k \leq n$).

This item is marked all or nothing: 10 points for a), otherwise 0 points.

General notes

- Alternative correct solutions receive full marks.
- Minor syntax errors in pseudocode, such as the form of the assignment arrow or a missing semicolon, are not penalised. Logic errors are.
- Partial points are awarded only where the criteria above say so.

Points summary

	Item 1	Item 2	Item 3	Item 4	Item 5	Item 6	Item 7	Item 8	Default	Total
Points	10	10	10	10	10	15	15	10	10	100